

Review

Hematophagous Tick Control in the South African Cattle Production System by Using Fossil Shell Flour as a Sustainable Solution: A Systematic Review

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Abstract: Ticks pose one of the key economic risks to the cattle industry globally, affecting productivity, health, and welfare. Over 80% of the cattle population around the globe is affected by tick infestation. Several tick control methods, including the use of chemical acaricides, herbal agents, and some complementary measures, like the use of old motor oil, disinfectant, pour-on, tick grease, pulling off, cutting, paraffin, and Jeyes fluid, have been used by farmers to alleviate the effects of hematophagous ticks. However, these strategies are often mistakenly employed and can harm animals. Likewise, these methods cannot be sustained because of their cost, environmental impact, flaws, and resistance. An abundantly available, cost-effective, environmentally friendly, and naturally occurring substance like fossil shell flour with no known side effects could present a sustainable solution. This review abridged the research and information on hematophagous tick control in South African cattle production using fossil shell flour. This is a systematic review of the published literature and catalogues. All available documented evidence on this topic was collated and synthesized through standardized methods of systematic review protocol. Different scientific studies and a few references from farmers' magazines published from 1941 to 2022 were reviewed. Out of 120 research papers downloaded, 98 were included and analyzed directly or indirectly regarding hematophagous tick control in cattle production and the use of fossil shell flour among livestock farmers. The advancement in ectoparasite control through fossil shell flour is a novel concept that needs to be explored for the benefit of all livestock farmers, hence this review. Fossil shell flour has been shown to have good insecticidal effectiveness against insects of animal and agricultural importance. We hereby recommend the exploration of FSF as an alternative tick control measure to the currently used acaricides to which ticks have developed resistance.

Keywords: acaricide resistance; ectoparasites; fossil shell flour; heat stress; tick infestation



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1. Introduction

Cattle production in South Africa is a vital agricultural activity, contributing to the beef, dairy, and leather industries. The country has a mix of commercial, small-scale, and communal farmers, with breeds adapted to diverse climates. Challenges include drought, disease management, and fluctuating market conditions [1]. Approximately 80% of the land in South Africa is primarily suitable for extensive livestock farming [1], of which 17% of the total farming area is communal land, which supports about 52% of the total cattle population [2]. Approximately 12.8 million cattle can be found in South Africa [3], of which a quarter (3.2 million) of these cattle is mainly produced in the Eastern Cape province by communal farmers [4]. However, the number of livestock marketed by communal farmers

in the Eastern Cape province is less than 10% of their herd size, compared with the 25–30% of that by commercial farmers [5]. Therefore, it is evident that communal farmers are challenged in occupying the formal market fully.

Ticks and tick-borne pathogens impact all livestock production systems but are most impactful on smaller-scale production systems, which are less able to absorb the cost of production losses, treatment, and control [6]. The adverse effects of ticks and tick-borne diseases have resulted in economic losses of over USD 33m annually in South Africa [4]. These challenges prohibit communal farmers from participating in the formal market since their animals would be downgraded because they do not meet the formal market standards. Consequently, an opportunity to increase household food security and export earnings while alleviating poverty is lost. Ticks and tick-borne diseases are the primary causes of substantial economic losses for cattle in most African countries, mainly from poor-resourced communal areas [7,8].

The decisions and implementation of tick and tick-borne pathogen control are primarily up to farmers [4], with acaricide use being the primary means of tick control, which reduces the incidence of tick-borne diseases. The high prevalence of ticks and tick-borne diseases in communal areas can be attributed to several factors: a lack of acaricidal resistance tests, the indiscriminate selling of acaricides, a lack of the rotation of acaricides, and a lack of farmer training on the use of acaricides [7]. Notably, two things need attention in addressing the issue of ticks and tick-borne diseases: On the one hand, tick control measures are farmer-dependent; thus, the affordability of acaricides and/or acaricide chemicals is an economic issue in communal farming. On the other hand, using the same acaricide over a long period and incorrectly using acaricide chemicals lead to increased tick resistance, resulting in a high tick population in non-endemic areas [9]. Therefore, the high prevalence of acaricide-resistant tick populations evokes a need for alternative tick control measures that will be cheap, readily available, and easy to apply. Thus, the use of fossil shell flour (FSF) is one of the promising alternative options.

Fossil shell flour, commonly known as diatomaceous earth (DE), consists of the remains of the microscopic single-celled plants (Phytoplankton) called diatoms found in oceans and lakes in many parts of the world [10]. These remains have long been mined from underwater beds or ancient dried lake bottoms for decades and have numerous industrial applications. Diatomaceous earth is mined, milled, and processed into various types for several uses. There are two main types of diatomaceous earth: food-grade and filter/non-food-grade [11]. Unlike filter-grade DE, food-grade DE is commonly used in the agricultural and food industries since it is considered safe for humans and animals [10,11]. Diatomaceous earth is a natural product that mechanically kills insects through dehydration by removing the waxy coating of their exoskeletons; hence, it is difficult for ticks to develop resistance against it [10,12–14].

Fossil shell flour is non-toxic, cheap, and readily available in many countries [10]. This makes affordability and availability two of the most significant advantages of FSF for its use by any farmer [10], especially communal farmers who are financially limited. FSF has recently been modified in livestock production as an additive for several uses [10]. It has been used as a feed additive, growth promoter, mycotoxin binder, water purifier, vaccine adjuvant, inert dust application in stored pest management, pesticide, and a natural source of silicon and anthelmintic [10]. Some farmers believe that feeding DE to livestock can kill internal and external parasites through the same mechanism. Nonetheless, this remains unproven, particularly for hematophagous ticks. Therefore, this article aims to review the use of FSF as an alternative tick control measure in cattle.

2. The Impacts of Ticks and Tick-Borne Pathogens on Livestock Production Systems

Ticks are the most critical group of parasitic arthropods, vectors of pathogens that affect domestic and wild animals [15]. Bovine babesiosis, anaplasmosis, theileriosis, and heartwater diseases are listed as notifiable by the World Organization for Animal Health as tick-borne diseases that cause transboundary livestock diseases [16]. Ticks that affect cattle around the world belong to two families: Ixodidae and Argasidae. Ixodidae, also known as hard ticks, includes all species from *Amblyomma*, *Dermacentor*, *Haemophysalis*, *Hyalomma*, *Ixodes*, and *Rhipicephalus*, while Argasidae (genus *Argas*) is a family of soft ticks which includes the *Ornithodoros* and *Otobius* ticks [17].

Ticks carry and transmit numerous protozoans and viral and bacterial pathogens. The most predominant infections associated with ticks infesting cattle in South Africa include theileriosis, babesiosis, anaplasmosis, Q-fever, and heartwater [18]. In [18], 130 ticks infesting cattle were sampled to identify tick pathogens. The infected ticks that were identified included *Rhipicephalus* (R.) *evertsi evertsi*, *R. appendiculatus*, *R. decoloratus*, and *Amblyomma*, while the most frequent pathogens identified were *Coxiella* (C.) *burnetti* (9.2%), *Rickettsia* spp. (7.7%), *Anaplasma marginale* (3.8%), *Theileria* (T.) *mutans* (3.1%), *T. taurotragi* (2.3%), and *Ehrlichia ruminantium* (1.5%), with the highest brown ear tick prevalence identified in ticks sampled from the Eastern Cape province.

In the Western Cape and Free State provinces, key tick-borne diseases in livestock include anaplasmosis, causing anemia and jaundice; redwater (babesiosis), leading to fever and hemoglobinuria; and heartwater, a fatal disease in ruminants marked by neurological symptoms [18].

On the other hand, gallsickness, heartwater, and redwater have been reported as the major tick-borne diseases that are predominant in livestock in the Eastern Cape province [19]. This clearly shows that communal farmers are still the most affected by ticks and tick-borne diseases, and an urgent sustainable intervention is needed in this sector.

3. Climate Change and Epidemiology of Ticks

The livestock sector contributes approximately 14.5% of anthropogenic greenhouse gas emissions [20]. The most significant greenhouse gases from livestock production are methane and nitrous oxide. Methane is mainly produced by enteric fermentation and manure storage and has a 28 times higher global warming effect than carbon dioxide. On the other hand, nitrous oxide also arises from manure storage, while organic and inorganic fertilizers have 265 times higher global warming potential than carbon dioxide [21]. This means that both these gas emissions significantly contribute to climate change, consequently affecting the epidemiology of ticks. Nuttall [22] reported the encroachment and spread of *I. scapularis* and *Lyme borreliosis* across much of Canada concomitant with a 2–3 °C increase in land surface temperature. Also, the northerly movement of *I. ricinus*, which is connected to tick-borne *encephalitis* (TBE), has been reported in northern Europe due to higher altitudes and rising temperatures in Central Europe [23]. In Africa, changes in the composition of ticks such as *Rhipicephalus microplus* and *Haemaphysalis longicornis* species are being reported due to increased heat tolerance [24]. This trend will continue in the next half decade, affecting the socioeconomic conditions linked to livestock and livelihood, especially in sub-Saharan Africa [22].

Climate change is defined as a long-term change in average weather patterns that have come to define the earth's local, regional, and global climates [25]. Some significant changes caused by climate change include warmer temperatures in temperate zones, altered precipitation patterns, the increased frequency and severity of extreme weather events (hurricanes or droughts), and rising sea levels [26,27]. Rising sea levels have

directly or indirectly affected the biology and ecology of many organisms on the planet [17]. Consequently, these climate variations have impacted ectoparasites' habits and biological cycles [26,28,29], including those of ticks [29].

Due to these climatic changes, ticks can evolve, adapt, and spread. This favours ticks' dynamics and population movement in different geographical areas [30]. This resulted in the presentation of relatively new infestations in some livestock areas or the diagnosis of diseases transmitted by these vectors, which were not common for certain latitudes in the past [31,32]. Climate change affects both domestic and wild animals [30,33], which influences the geographical distribution of ticks, their infestation, and the diseases they transmit in non-endemic areas [34]. Thus, mitigation strategies to reduce gas emissions from the livestock sector are needed to limit the epidemiology of ticks.

4. Previous and Current Tick Control Measures

Acaricide application through dipping has always been the major tick control measure used by communal farmers in most communal areas of South Africa. Nonetheless, these farmers have also used complementary tick control measures, including old motor oil, disinfectant, pour-on, tick grease, pulling off, cutting, paraffin, and Jeyes fluid, with the last of these being the least preferred method [35]. Recently, tick control has been achieved by using chemical treatments (acaricides and insecticides), which have contributed significantly to improving the productivity and welfare of animals [17]. Nonetheless, these therapeutic methodologies' intensive, frequent, and inappropriate use has resulted in acaricidal resistance in ticks [36–39]. Promising strategies for tick control include innovative approaches like the TickBot robot and devices aimed at reservoir hosts such as deer. Additionally, future directions involve developing vaccines that target tick antigens and implementing translational biotechnological methods to manage tick populations [35].

Tick resistance has been reported for almost all chemical acaricides [39,40]. Consequently, this has created a worsened chemical control problem, which also has consequences that include environmental and food contamination by secondary chemical metabolites, the spread of ticks to free zones, restrictions on cattle export, and an increase in diseases transmitted by these parasites [41–43]. Research has shown that dependence on these chemical products as the only control measure on ticks is neither economically nor ecologically sustainable [17]. Considering novel tick control measures will yield sustainable cattle production. Thus, exploring alternative methods like the use of FSF could yield sustainable cattle production.

5. Acaricide Resistance of Livestock Ticks in South Africa

Acaricide resistance among livestock ticks is a significant concern in South Africa, impacting the efficacy of tick control measures and posing challenges to animal health. The primary tick species affecting cattle in the region include *Rhipicephalus (Boophilus) decoloratus* and *Rhipicephalus (Boophilus) microplus*, both known vectors of diseases such as babesiosis and anaplasmosis. [44]. Commonly used acaricides employed in South Africa include organophosphates, synthetic pyrethroids, Amidines, and Macrocyclic Lactones [45]. This is the worst-case scenario, particularly for poor-resourced farmers who use these chemicals with limited knowledge.

Acaricide resistance can be defined as the case when ticks develop abilities to increase survival despite the use of chemicals to control them significantly [46]. This is an evolutionary adaptation that was caused by selecting specific heritable traits in a tick population resulting from intensive acaricide exposure [45]. Most acaricides work on ticks' nervous system [47]. Nonetheless, the mechanisms by which ticks resist acaricides are metabolic detoxification and point mutations at target sites, which prevent the action of

acaricides [48,49]. As a result, tick acaricide resistance has become a topic of research interest worldwide [50–53], especially owing to major challenges and expenses in developing new acaricides and producing cattle breeds that are tick-resistant [54–57].

A study in the Eastern Cape province assessed *R. decoloratus* ticks from cattle and found its resistance to multiple acaricides, including synthetic pyrethroids and organophosphates. This resistance complicates effective tick control in the region [58,59]. A similar study in the same region highlighted the competitive displacement between *R. decoloratus* and *R. microplus* on commercial farms, noting that *R. microplus* exhibited higher resistance levels to commonly used acaricides [60,61]. Research focusing on the evolution of acaricide resistance in *R. decoloratus* emphasized the role of genetic mutations and selection pressure from prolonged acaricide use, leading to decreased susceptibility [62]. Therefore, there is a need for mitigation strategies through integrated control strategies that include the use of FSF to limit the use of chemicals.

6. Physical and Chemical Properties of Natural Fossil Shell Flour and Chemical Composition of FSF Deposits Found in South Africa

Fossil shell flour is a white powder that occurs naturally as a soft sedimentary rock that contains approximately 80–90% silica [63,64] with a size ranging between 0.75 μ m and 1500 μ m [65]. It is formed naturally due to the accumulation of the silica cell walls of dead, microscopic, single-celled, aquatic algae called diatoms [63]. It has distinctive properties such as its permeability (0.1–10 MD), porosity (35–65%), small particle size, low density, thermal conductivity [66], large surface area [67], and low specific gravity, and it is highly siliceous and relatively inert [68]. These properties make FSF quite interesting among naturally occurring materials and useful for many industrial purposes [68–70]. Furthermore, the extracted FSF from geological deposits may contain diverse metal oxides and organic matter associated with the dominant silica content (SiO_2). The most abundant oxides are Al_2O_3 , Fe_2O_3 , CaO , MgO , K_2O , Na_2O , and P_2O_5 .

This review summarizes the chemical composition of FSF deposits found in South Africa (Table 1). In [10], the authors explored the chemical composition of FSF deposits from several other sources/countries.

Table 1. Chemical composition of FSF deposits found in South Africa.

Chemical Content	% Weight
silicon dioxide	81.65
aluminum oxide	0.89
iron(III) oxide	-
calcium oxide	15.69
magnesium oxide	1.08
manganese oxide	-
potassium oxide	0.28
sodium oxide	-
titanium dioxide	-
phosphorus pentoxide	0.126
loss of ignition	-

Source: [71].

7. Fossil Shell Flour as an Insecticide

Fossil shell flour (FSF), also known as diatomaceous earth (DE), has been reported as a very effective natural insecticide [10,12,72,73]. Diatomaceous earth is a light dust which contains porous particles with abrasive properties and the ability to absorb lipids [12]. Rough-bodied insects easily pick up these particles; they damage the exoskeletons of

these insects through hydrocarbon absorption and abrasion, which make their cuticles permeable to water, resulting in water loss and death by dehydration [12,74]. It was, therefore, reported that any DE with high absorbing capacity has the potential to be used as an insecticide [10,12].

Beyond FSF's capacity to absorb approximately three or more times the particle mass, the size of the particles, the uniformity, shape, pH, and purity of the formulation affect its efficacy [12]. For instance, DE formulations can be more effective as insecticides if they have a pH < 8.5, have highly pure amorphous silica with uniformly sized diameter particles (<10 µm), contain less than 1% crystalline silica, and have the lowest possible number of clay particles or impurities [12,75–77]. By containing these particles, DE works purely in a physical and/or mechanical manner and is believed to kill external parasites in the same manner by lacerating the vulnerable parts of parasites and killing them by dehydration.

Research has also been conducted on the efficacy of DE on other insects such as textile pests, ants, bedbugs, termites, several caterpillars in agriculture, earwigs, potato beetles, June beetles, fleas, silverfish, poultry mites, centipedes, snails, pillbugs, and ticks [72,78–81]. From these studies, it was concluded that insects' sensitivity to DE differs due to their anatomy and physiology. For instance, smaller insects are more sensitive to DE since they lose greater amounts of water from their bodies, while insects with a rough or hairy body surface collect more of the DE particles per unit area, resulting in greater cuticle damage [12,82]. Also, insects with a thin epicuticle are more sensitive than those with a thicker epicuticle [83], while insects with softer wax coats are more sensitive than those with a hard wax coat [74].

Furthermore, Dawson [84] reported a reduction in the bird flea population (*Ceratophyllus* ideas) and numerous blow fly species (*Protocalliphora* spp.) when DE is used. Ikusika [10] attributed the ability of FSF to reduce the flea population to the body sizes of fleas that are smaller and, therefore, more susceptible to dehydration and the abrasive action of FSF. Recent studies by Martin and Mullens [85] and Murillo [86] also confirmed that FSF can suppress the activities of Northern fowl mites (*Acari: Macronyssidae*). At the same time, Kilpinen and Steenberg [87] reported the use of FSF as being one of Europe's most used substitute control methods for poultry red mites (*Dermanyssus gallinae*) since it kills the target host by dehydration. Most interestingly, Flanders [88] previously highlighted that insects that can recover water loss from their bodies, such as sucking insects and mites, are more resistant than those that should metabolize water from their food. Nonetheless, Ebeling [74] argued that resistance is unlikely since the use of DE only acts as a physical control method. Similarly, Dawson [84], Isabirye et al. [89], and Mthi et al. [90] all reported that insects and ectoparasites are not prone to resistance to diatomaceous earth.

Although the action of FSF on ectoparasites is reported to be similar to that of endoparasites, it is still unclear. Nonetheless, FSF is known to be rich in trace elements; thus, McLean et al. [91] suggested that animals enhanced nutritional status may enable them to cope with parasite burdens. For instance, livestock fed with DE have notably shinier coats due to the mineralization effects of natural food-grade DE. Natural food-grade DE contains 15 trace minerals, including magnesium, calcium, potassium, zinc, copper, iron, selenium, sodium, and phosphorus, which improve livestock's overall health and production. Particularly, because the fossilized algae from which DE is extracted contains high amounts of silica, DE is a form of silica known to give cattle strong hoofs and healthier and shinier coats.

Some studies have indicated an interesting possibility of tick resistance attributed to coat characteristics; this relationship is still unclear. For instance, cattle with short, slick hair coats have been reported to be tick-resistant [92], while Olson et al. [93] attributed this to the dominant slick hair gene. Bonsma [94] and Hupp [95] indicated that cattle with longer hair are known to have high levels of tick infestation. Peters et al. [96] also

confirmed that the coat type significantly controls ectoparasites. Furthermore, Fraga et al. [97] reported a high level of tick infestation in cattle with thicker hair coats. The authors further highlighted that thick hair allows for suitable temperature and humidity conditions conducive to parasite development.

Ibelli et al. [98] compared different genetic groups in terms of tick infestation; the authors concluded that although traits influencing an animal's ability to resist ticks vary with genetic groups, the coat thickness and sample weight of hair were positively correlated with tick infestation. Furthermore, cattle with short and shiny coats are known for heat tolerance, i.e., reduced heat stress [99]; hence, Fraga et al. [98] further assumed that heat stress increases glucocorticoid levels in the bloodstream, affecting antiallergic and anti-inflammatory effects, thus favouring tick infestation.

Based on the reviewed research in this article, we can suggest that there is a linkage between heat stress and tick infestation. Although this relationship is unclear, mitigation strategies to control tick infestation may be achieved by improving cattle coat characteristics using FSF. Although no scientific evidence supports the idea that FSF improves cattle characteristics, several farmers have regularly noted shinier coats in their animals fed food-grade diatomaceous earth [100]. Ikusika et al. [10] further suggested that the residual mineral content of FSF may also give animals a shinier coat. This, therefore, evokes a need for researchers to authenticate these claims to determine the impact of FSF on coat characteristics and its efficacy as an insecticide.

8. Availability, Accessibility, Research, and Development of FSF in South Africa

Several countries are actively involved in the mining, milling, and processing of FSF. Kilpinen and Steenberg [87] reported that DE from different sources has different strengths against parasites, especially in poultry. This evokes a need to research and develop diatom deposits in South Africa. Table 2 shows the 2019 global production of FSF in major countries. Table 3 shows some comparisons between FSF and acaricides.

Table 2. Global FSF production.

Country	% Production
United States of America (USA)	34%
China	15%
Denmark	15%
Turkey	6%
Republic of Korea	5%
Peru	4%
Mexico	3%

Source: [101].

Table 3. Comparison between fossil shell flour and acaricides.

FSF	Acaricide
Environmentally friendly	Most of them have adverse effects on the ecosystem.
Slow action	Quick action.
Insects are unlikely to develop resistance against it	Insects often develop resistance against it.
Cheap and readily available for commercial and communal farmers	It is often expensive for communal farmers.
Organic products	Inorganic products.

Source: [10,34].

In South Africa, FSF is mined from seven different freshwater sources in the Olifan-shoek area [100]. Since FSF is famously known to be cheap, readily available, effective against insects, and safe, all commercial or communal farmers can use it. As a result, numerous South African farmers are already using FSF with great success, especially in the Northern Cape [24]. However, more evidence-based scientific work is required to explore various benefits and applications of fossil shell flour, such as its use in controlling hematophagous ticks, which has not been explored. Also, more research that will provide information on the optimal feeding level response is crucial to make FSF more effective in livestock production practices among all farmers and to confirm the efficacy of the used dosages in current peer-reviewed work.

However, because fossil shell flour consists of nanoparticles and has low density, it can easily evaporate into air and be inhaled. After inhalation, these nanoparticles have been observed to quickly pass through the alveolar-capillary barrier and enter the systemic circulation, reaching different organs [102]. According to [103], diatomaceous earth, when applied to the skin, has the potential to cause wounds or skin damage. These challenges can be circumvented by applying FSF through encapsulation in biodegradable carriers such as chitosan or liposomes, as described by [104]. Also, mixing FSF with high-density carriers such as bentonite clay or kaolin improves its dispersibility [105]. Transforming FSF and nanoparticles into a nanoemulsion or aqueous suspension improves adherence and spreadability. Nanoemulsions containing essential oils like citronella or neem enhance tick repellence while ensuring even distribution [106]. Another way of resolving the issue of the use of FSF nanoparticles is the addition of bioadhesive agents such as glycerin or natural gums that improve the adhesion of FSF to livestock hair and skin [107]. Therefore, in addition to these non-beneficial effects of DE, the mode of application of DE should be carefully selected so that the benefits of controlling ticks in cattle using DE can be successfully provided.

9. Conclusions

Fossil shell flour has been shown to have good insecticidal effectiveness against insects of animal and agricultural importance. Few studies and/or little research has focused on FSF as a dewormer, i.e., to control endoparasites. Nonetheless, following the economic and health importance of ectoparasites, particularly hematophagous ticks, we recommend the exploration of the use of FSF as an alternative tick control measure to the currently used acaricides to which ticks have developed resistance. The formulation/choice of FSF is critical to the success of the use of FSF as a tick control method in cattle. Thus, researchers need to investigate the efficacy of different sources of FSF and their dosages as an alternative to anthelmintics.

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